

THOEN, Thailand

WMO: 483240

Lat: 17.6364N

Lon: 99.2447E

Elev: 192

StdP: 99.04

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	14.5	8.1	6.9	26.8	9.9	7.8	26.1	5.8	30.8	4.7	29.1	0.6	N/A	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.3	39.8	23.4	38.4	23.4	37.2	23.3	27.1	32.3	26.7	31.8	26.3	31.4	2.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.6	29.1	25.5	21.2	28.7	25.1	20.7	28.2	86.6	32.1	84.9	31.9	83.5	31.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.4	3.5	DB	9.8	41.5	1.7	1.2	8.6	42.3	7.5	43.0	6.6	43.7	5.3	44.6	
			WB	9.1	28.0	1.8	0.4	7.8	28.3	6.8	28.5	5.8	28.7	4.5	29.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	23.5	26.2	29.1	31.2	30.0	28.5	28.3	28.0	28.1	27.2	26.2	23.6
	DBStd	2.87	2.46	1.89	1.70	2.22	2.17	1.35	1.37	1.01	1.11	1.29	1.69	2.42
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	2	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6387	418	453	593	636	619	557	568	559	542	534	486	421
	CDD18.3	3348	162	220	334	386	361	307	310	301	292	276	236	163
	CDH23.3	36153	1458	2531	4409	5327	4323	3329	3294	2951	2805	2392	1993	1341
Wind	CDH26.7	17140	604	1351	2639	3231	2248	1345	1304	1069	1070	926	841	514
	WSAvg	1.0	0.9	1.2	1.4	1.4	1.2	1.0	1.1	0.9	0.8	0.7	0.7	0.8
	PrecAvg	1209	10	12	30	63	189	153	141	185	243	149	28	9
	PrecMax	1754	54	71	160	161	382	264	259	263	365	275	112	70
	PrecMin	860	0	0	0	7	33	47	81	102	131	34	0	0
	PrecStd	211	14	19	38	41	108	56	46	54	69	67	30	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	37.4	40.2	42.0	41.0	37.0	36.8	34.5	35.2	35.0	35.1	33.9
		MCWB	21.1	20.7	21.8	23.0	23.9	25.3	24.8	25.8	26.4	25.9	24.4	22.9
	2%	DB	32.9	36.1	38.8	40.5	38.7	34.8	34.6	33.3	33.9	33.7	33.5	32.6
		MCWB	21.2	21.0	22.1	23.4	24.8	25.7	25.0	25.7	26.0	25.7	24.3	22.7
	5%	DB	31.7	35.0	37.7	39.0	36.8	33.5	33.0	32.3	32.9	32.5	32.4	31.4
		MCWB	21.2	21.2	21.6	23.8	25.2	25.7	25.3	25.5	25.8	25.6	24.2	22.5
	10%	DB	30.3	33.6	36.5	37.5	35.2	32.2	31.8	31.4	31.7	31.3	31.2	29.8
		MCWB	21.3	21.4	21.8	24.1	25.4	25.6	25.2	25.3	25.6	25.4	24.1	22.1
	0.4%	WB	24.7	25.0	25.4	26.7	27.5	27.3	26.8	27.2	27.5	27.1	26.6	25.0
		MCDB	29.9	32.0	34.2	34.7	33.7	32.1	31.4	31.9	32.9	32.1	31.8	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.8	23.9	24.8	26.0	26.7	26.7	26.2	26.5	26.8	26.5	25.7	24.2
		MCDB	28.7	31.5	33.0	34.5	32.5	31.4	30.8	31.0	31.8	31.3	30.8	29.8
	5%	WB	22.9	23.2	24.2	25.6	26.2	26.3	25.9	26.1	26.4	26.1	25.2	23.5
		MCDB	27.8	30.7	32.5	33.8	32.3	31.0	30.4	30.5	31.0	30.8	30.1	28.6
	10%	WB	22.2	22.6	23.7	25.2	25.9	26.0	25.6	25.7	26.0	25.7	24.7	22.7
		MCDB	27.5	29.8	31.9	33.1	31.9	30.6	29.9	29.8	30.3	30.0	29.2	27.6
	5% DB	MDBR	13.8	15.9	15.1	12.3	9.9	7.6	7.4	7.5	8.0	8.4	10.6	12.4
		MCDBR	16.0	17.8	17.4	14.4	12.7	9.7	9.3	8.9	9.8	10.5	12.3	14.1
		MCWBR	5.3	4.9	4.0	2.8	2.4	2.5	2.4	2.6	2.9	3.6	4.1	4.8
		MCWBR	11.8	13.6	13.3	11.7	9.9	7.9	7.6	7.9	8.8	8.8	9.9	11.3
Clear-Sky Solar Irradiance	taub	0.456	0.542	0.666	0.637	0.498	0.438	0.427	0.436	0.472	0.535	0.458	0.453	
	taud	2.152	1.933	1.665	1.782	2.198	2.394	2.423	2.380	2.264	2.039	2.229	2.184	
	Ebn at Noon	822	768	690	714	811	853	863	862	833	765	816	812	
	Edh at Noon	145	189	254	227	148	120	117	123	138	167	133	137	
All-Sky Solar Radiation	RadAvg	4.66	5.32	5.58	6.05	5.47	4.75	4.44	4.39	4.58	4.59	4.71	4.44	
	RadStd	0.23	0.22	0.34	0.34	0.49	0.40	0.35	0.28	0.25	0.27	0.29	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	+0.26	N/A	N/A	N/A	N/A	+0.13	N/A	N/A	+121	+123

Nomenclature: See separate page